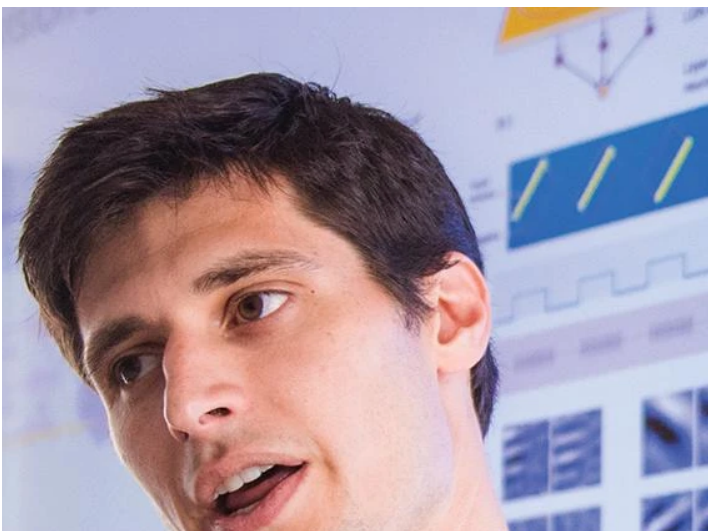


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The Personal Mission Behind One MD-PhD'S Research



The motivations behind the research of Sam Finlayson

Artificial Intelligence Issue

 by Randall Fox



3 minute read



Profile

Sam Finlayson's desire to become a physician-scientist is rooted in hope and heartbreak. A fifth-year student in the Harvard/MIT MD-PhD Program who is working on his doctorate in systems biology in the Department of Biomedical Informatics in the Blavatnik Institute at HMS, Finlayson overcame severe health problems to get where he is today.

While a newborn, he contracted meningitis, which progressed to encephalitis and precipitated a years-long nightmare that included seizures, strokes, oxygen support, a surgical error that resulted in intravenous feeding, and developmental delays.

But, he says, it's not his personal health hurdles that motivate him to pursue the research he undertakes today. It's his sister, Kate, and the health problems she faced.

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“I have tried to craft a career in which I could provide care to patients according to the best knowledge we have, while also nudging the boundary of our capacity a little further.”

As a child, Kate had a brain hemorrhage that resulted in hydrocephalus. When she was in middle school, a shunt implanted in her brain to divert excess cerebrospinal fluid failed, requiring surgery that caused complications. After more than one hundred brain surgeries and years of hospitalizations, she died in 2010 at age twenty-six.

“My sister and I and our whole family have really experienced the very best and the very worst of medicine,” Finlayson says. “Since early high school, I have tried to craft a career in

which I could provide care to patients according to the best knowledge we have, while also nudging the boundary of our capacity a little further.”

Finlayson now works in the lab of his mentor, Isaac Kohane, chair of the biomedical informatics department, and focuses on using machine learning to find treatments for rare diseases.

“We have compiled some incredible data sets measuring the effect of small-molecule drugs on gene expression in cells,” says Finlayson. “I’m training large neural networks to learn the patterns in these data sets and others like them.”

“The goal,” he adds, “is to apply these same algorithms to data collected from hard-to-treat diseases and to identify drugs that could work against them.”

Ten years ago, he and his brother created Team Hydro, a group that swims to raise money for hydrocephalus research. Since then, the team has raised more than \$700,000 and sponsored ten sizable research grants. Swims across Boston Harbor have featured many of Finlayson’s HMS classmates, as well as Kohane.

“The growth of Team Hydro is a testament to both the legacy of hope and perseverance that Kate left behind and the tenacity of my brother and my parents,” Finlayson says.

Image: Gretchen Ertl

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